An open-chord transfer for same-disorder curvature

Local source geometry and the Slater selected representation

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Abstract

The fixed-Slater composite source formula is reduced to an explicit recursion on open endpoints and two kinds of links. A normalized two-label collision can retain only one quadratic mark, a pair of nested quadratic marks, or linear marks on the same Gaussian edge. This gives a complete source classification, including arbitrary selected spectators. The transfer sums all fixed mixed words in the Hamiltonian, size fluctuation, curvature and selected operators without storing closed diagram history. Its reachable module carries the positive Gram form inherited from the microscopic limit. The auxiliary configurations themselves are not an orthonormal basis.

1 Definitions and the source formula used as input

Use standard CAR, the overlapping Hermitian complex Gaussian p-body Hamiltonian, one common disorder J, and a fixed half-filled Slater vector O. The state is annealed, $\varphi_N = \mathbb{E}\langle O| \cdot |O\rangle$. Take $m = N/2$, $\lambda_N = p^2/N \to \lambda > 0$, at fixed word length. Let $C_Q = \mathbb{E}H^2$ and C_m be its scalar on charge m. Define

$$\begin{aligned} DX &= QX - \sum_i c_i^\dagger X c_i, & \mathcal{L}X &= \tfrac{1}{2}(NX - \sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger)), \\ h_N &= H_m/\sqrt{C_m}, & W_N &= (p - \mathcal{L} - \lambda_N)H|_m/\sqrt{C_m}, \\ j_N &= C_m^{-1}(2p - D - \kappa_N)(H^2 - C_Q)|_m, & \kappa_N &= 2\lambda_N \frac{1-q_N}{1+q_N}, \quad q_N = e^{-4\lambda_N}. \end{aligned} \tag{1}$$

Loss maps act on full Fock space before sector restriction. The denominator is the fixed target-sector constant, not C_Q inside a loss map. Use dimensionless energy and $\hbar = 1$. In the limit put

$$q = e^{-4\lambda}, \quad \gamma = e^{-\lambda}, \quad \kappa = 2\lambda\frac{1-q}{1+q}, \quad W = \sqrt{\lambda}\,u, \quad v = \varphi(j^2) = 4\lambda + \frac{16\lambda^2 q}{1+q}. \tag{2}$$

$M^\dagger$ is the normalized occupied-to-empty block of the same original disorder; $\varphi(MM^\dagger) = 1$.

The preceding project stage expands a j into two adjacent H slots with a whole-block D source, excluding its internal Gaussian pair. W is one H slot with an $\mathcal{L}$ source. An edge is unrestricted (G), or forward selected (F). Backward edges vanish in this limit. For a compatible pairing π, write b for mixed H/selected edges and ν for crossings plus nestings whose inner edge is F. Its weight is

$$\gamma^b[z_{\mathrm{all}}] \exp\left(\lambda K_\pi(z) - \lambda \sum_W z_a - \kappa \sum_j z_a\right), \qquad K_\pi(0) = -4\nu. \tag{3}$$

All source variables are commuting and square-zero. We reorganize this already established fixed-word limit; no new N-limit is taken.

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2 The complete local source classification

Write $\pi(a)$ for the partner of endpoint a. A quadratic mark at $(a, a+1)$ meets two different edges e, f. Let $\epsilon = 1 - 2\mathbf{1}_{e \text{ crosses } f}$. Define α_a as follows:

$$\alpha_a = \begin{cases} 2\epsilon\mathbf{1}_{e_* \text{ is } G}, & \pi(a), \pi(a+1) > a+1, \ e_* \text{ has the smaller right endpoint}, \\ 2\epsilon\mathbf{1}_{e_* \text{ is } G}, & \pi(a), \pi(a+1) < a, \ e_* \text{ has the larger left endpoint}, \\ 2\epsilon, & \text{the partners lie on opposite sides.} \end{cases} \tag{4}$$

Lemma 1. *The nonconstant coefficients of K_π are precisely these:*

- *$K_W = 1$ on a G edge and 0 on an F edge; $K_{WW'} = 1$ if both marks are endpoints of the same G edge.*
- *$K_j = \alpha_a$. Two quadratic marks at $(a, a+1)$ and $(c, c+1)$, $a < c$, have $K_{jj'} = 4$ exactly when $\pi(a) = c+1$, $\pi(a+1) = c$; otherwise it is zero.*
- *Every mixed K_{Wj} and every coefficient of source order at least three vanishes.*

Proof. The normalized collision expansion sums memberships containing two labels. A site operator at an endpoint with neither label is I. At a marked block containing just one label it is odd; both loss maps multiply it by $1/2$. Its factor $1 - z/2$ cancels the corresponding singleton denominator, even when other marked blocks are present. The Jordan–Wigner sign for that membership involves only its two incident edges. Spectator edges therefore disappear identically.

If the labels are P, Q of one edge, a quadratic block containing one endpoint has $D(n) = n$, canceled by its two singleton factors $(1 - z/2)^2 = 1 - z$. For an $\mathcal{L}$ mark, $\mathcal{L}(n) = n - I/2$. Summing the empty and occupied matrix elements with the factor 2 in K leaves the coefficients $1, 1, 1$ for the two possible linear marks and their product. An F edge forbids this double membership, so all these coefficients vanish there.

For labels on different edges, every linear mark cancels. A surviving D mark must contain both incidences. Each edge has two endpoints, so at most two such marks survive. Evaluation of the four choices $P_eP_f, P_eQ_f, Q_eP_f, Q_eQ_f$ in the two site states gives the table below. Two surviving marks consume all four endpoints: the same calculation gives 4 for the nested connection and 0 for the crossed one. Both edges then have H endpoints, hence are G. □

After deleting spectators, label endpoints $0, 1, 2, 3$ and order edges by their left endpoints. The complete one-mark table is

Mark start	Pairing	GG	GF	FG	FF
0	(02)(13)	−2	−2	0	0
0	(03)(12)	2	0	2	0
1	(01)(23)	2	2	2	2
1	(02)(13)	−2	−2	−2	−2
2	(02)(13)	−2	0	−2	0
2	(03)(12)	2	0	2	0

3 Open configurations and single endpoints

The cut-diagram transfer is an established method [1, Sec. 2.3]; state-adapted dressed operators and their GNS interpretation have related precedents [2, Secs. 3.2–4]. Here the additional information comes from the common-disorder sources.

Scan from right to left. A configuration $(s, \mathcal{C})$ consists of an ordered list s of open right endpoints, oldest first, and disjoint two-endpoint links $\mathcal{C}$. The five endpoint colors are

Color	Edge type	Right endpoint
H	G	ordinary Hamiltonian slot
h	F	ordinary Hamiltonian slot
U	G	linear marked slot W
w	F	linear marked slot W
P	F	$M^\dagger$

Specifying G/F when opening a Hamiltonian slot is a sum over future compatible closures, not an additional factor. Every completed edge has exactly one such type. The vacuum is the empty configuration.

Opening H gives the two branches H, h; opening W gives U, w; $M^\dagger$ only appends P, and M cannot open. With n open endpoints and a zero-based position i, the geometric factor for closing is

$$\omega_i = \begin{cases} q^{n-1-i}, & G, \\ q^{n-1}, & F. \end{cases} \tag{5}$$

There are $n-1-i$ younger open edges, each crossing the closing edge. When the closing edge is F, each of the i older open edges also forms an inner-F nesting. This proves (5). Multiply it by the following endpoint coefficient; blanks are zero:

Closing insertion	H	h	U	w	P
H	1				γ
W			λ		$-\lambda\gamma$
M		γ		$-\lambda\gamma$	1

Indeed an unpaired W on a G edge has coefficient $\lambda(K_W - 1) = 0$. A pair of W endpoints contributes λ; a selected W endpoint contributes $-\lambda$. This already sums all linear sources, including their centering terms.

A link of type S joins the two still-open edges of a singleton-assigned quadratic mark. At its first closure multiply by

$$2\lambda\epsilon\mathbf{1}_G - \kappa, \qquad \epsilon = \begin{cases} +1, & \text{younger linked endpoint closes first,} \\ -1, & \text{older linked endpoint closes first.} \end{cases} \tag{6}$$

Delete that link; the other endpoint is now ordinary. A link of type R is reserved for a quadratic source pair and cannot close at an individual endpoint. Relabel surviving positions after every deletion.

4 The explicit quadratic insertion

Expand the coefficient of all quadratic sources in (3) into singleton blocks of weight $\lambda\alpha - \kappa$ and pair blocks of weight 4λ. This is the finite set-partition expansion of the square-zero exponential; no higher source block exists.

For a singleton-assigned j, process its right slot and then its left slot as H. The two slots cannot pair with one another. There are four possibilities, with all ordinary closure factors from the previous page:

OO. Open both endpoints (each has its G/F branches), join them by an S link, and defer the singleton weight to its first closure.

OC. Open the right endpoint, close an endpoint that was already open before this insertion, and multiply by $-2\lambda - \kappa$. The two incident edges cross.

CO. Close an old endpoint, open the left endpoint, and multiply by $2\lambda - \kappa$. The two incident edges are disjoint.

CC. Close two distinct old endpoints with original indices i, k, in that order. Put $\epsilon = +1$ if $i > k$ and -1 otherwise. The younger original endpoint has index $\max(i, k)$. Multiply by

$$2\lambda\epsilon\mathbf{1}_{s_{\max(i,k)} \text{ is } G} - \kappa. \tag{7}$$

Any old S link encountered during either closure is resolved as well. Old R endpoints are unavailable to these singleton branches.

The remaining branches assign the current quadratic mark to a source pair:

Pair opening. Append two H endpoints and join them by an R link, with coefficient one.

Pair closing. Choose an existing R link (a, b), $a < b$. Close b and then a, delete the link, and multiply by

$$4\lambda\, q^{(n-1-b)+(n-2-a)}. \tag{8}$$

The two edges are nested and unrestricted. Their crossings with other open edges are precisely the two exponents in (8).

The links are always disjoint: an S link is created only on two new endpoints and is erased when either is removed; an R link is created and removed as a whole. Consequently the record stores no information about already closed edges. Configurations with the same ordered colors and links are merged by adding their coefficients.

For a prescribed finite word, one may discard configurations that cannot close in its unprocessed prefix. The implementation checks the remaining number of slots, M closures for h, w, W closures for U, available H slots for H, and j insertions for R links. These are necessary conditions only; using them cannot discard a valid completed path. This is exact pruning, not a numerical cutoff or a bound on worst-case computational complexity.

5 Equivalence and the physical Gram form

Let T_A denote the preceding recursion for an insertion A, and ε the coefficient of the empty configuration. For a word $w = A_1 \cdots A_r$, set $T_w = T_{A_1} \cdots T_{A_r}$.

Theorem 1. *For every fixed word in $h, W, j, M, M^\dagger$,*

$$\varphi(w) = \varepsilon T_w \emptyset. \tag{9}$$

The reachable span $V = \mathrm{span}\{T_P\emptyset\}$ has a well-defined positive semidefinite form

$$G(T_P\emptyset, T_Q\emptyset) = \varphi(P^\dagger Q). \tag{10}$$

Its null quotient is the joint polynomial GNS module, and the induced actions satisfy $T_A^* = T_{A^\dagger}$ on that core.

Proof. Decorate a compatible forward pairing with one term of its finite source-partition expansion. Each j is either a singleton or paired with another j by the nested two-edge connection of Lemma 1. Scan the decorated pairing. Ordinary closures supply $\gamma^b q^\nu$ and the linear-source weights. A singleton quadratic mark follows exactly one OO, OC, CO or CC branch; the formula for α gives respectively (6), the two split weights, or (7). A source pair follows a pair opening and (8). Conversely each completed path specifies this decorated pairing uniquely. Thus the finite sums coincide. Merging identical open records preserves all future weights, since those records contain every unresolved source and edge. This proves (9).

If $T_P\emptyset = 0$, then for every polynomial R, $\varphi(RP) = \varepsilon T_R T_P \emptyset = 0$. Hence changes of polynomial representative do not affect (10) in its second slot, or in its first slot by Hermiticity. Positivity follows from the established finite-N limit of $\mathbb{E}\,\|PO\|^2$. The form is the usual positive moment form, whose null space is a left ideal by Cauchy–Schwarz. Its quotient and adjoint relations therefore give exactly the polynomial GNS construction. □

This argument gives the metric on the *reachable* module. It does not assign a Euclidean metric to formal open configurations, nor assert that every such configuration is individually reachable. In practice a word-basis Gram matrix is computed by applying the same recurrence to $P_a^\dagger P_b$.

The previous polynomial-density theorem bounds the covariant algebra on this entire GNS space. In particular

$$B_h = \frac{2}{\sqrt{1-q}}, \quad \|W\| \le \sqrt{\lambda} B_h, \quad \|j\| \le \frac{2\sqrt{v}}{\sqrt{1-q^4}}. \tag{11}$$

For $S_K(t) = \sum_{r=0}^{K} (ith)^r / r!$, the limiting-space estimate is

$$\left\| (e^{ith} - S_K(t))\psi \right\| \le \|\psi\|\; e^{|t|B_h} \frac{(|t|B_h)^{K+1}}{(K+1)!}. \tag{12}$$

It provides controlled limiting covariant time calculations between fixed preparations; it does not by itself exchange finite-N, large-time and nonlinear-pinching limits.

6 Exact checks, scope and the next calculation

The fresh checker uses rational arithmetic and imports only pure functions from preceding stages. Its first completed scientific output records:

Comparison	Count
Full local CAR source exponent versus geometric rule	2774
Selected source moments versus transfer (497 words, two points)	994
Old $H/U/P$ transfer: all six-letter $H, W, M, M^\dagger$ words	4096
Positive exact 14×14 Gram LDL decompositions	2
Unpruned versus pruned vacuum coefficient checks	5

The two formal points are $(\gamma, \lambda) = (1/2, 1/3), (2/3, 1/4)$. They test rational identities off the physical curve $\gamma = e^{-\lambda}$. Physical positivity is inherited from the microscopic theorem, not inferred from these finite tests.

Longer examples provide an actual use of the transfer:

Word	Expanded endpoints	Peak retained states
h^{24}	24	7
j^8	16	21
$jh^2jW^2jh^2j$	14	10
$Mjh^4jM^\dagger$	10	18
$M^2jh^2j(M^\dagger)^2$	10	22

All are exact at the first formal point. The first reference is the classical Touchard–Riordan formula [3],

$$\varphi(h^{2n}) = \frac{1}{(1-q)^n} \sum_{k=0}^{n} (-1)^k \left[\binom{2n}{n-k} - \binom{2n}{n-k-1} \right] q^{k(k+1)/2}. \tag{13}$$

The next two use the preceding tracial mixed-q law; the selected examples use the original full source engine. A preliminary checker was interrupted after mistakenly choosing full pairing enumeration for the h^{24} reference. It had written no scientific output. Replacing that reference by the displayed classical sum allowed the first complete run to finish. The interruption is retained in the diagnostics.

The new result is an arbitrary-fixed-word computational representation with a physical Gram quotient. It does not classify the full selected operator algebra, prove compactness of M in this enlarged space, or provide the spatial and mass generators of de Sitter holography. The old bridge for actual curvature $a_N = (1 - \mathcal{P}_H)x_N$, $x = j/\sqrt{v}$, continues to apply to covariant blocks between fixed polynomial preparations. General interlacing of nonlinear pinched blocks remains a separate limit problem.

The next useful physical calculation is the full restriction of a single selected excitation to the three-field covariant algebra, followed by higher bilocal descendants. It can now be tested with explicit ordered mixed moments while retaining the same original disorder, before choosing a spectral or bulk interpretation.

References

[1] M. Berkooz, M. Isachenkov, V. Narovlansky and G. Torrents, *Towards a full solution of the large N double-scaled SYK model*, arXiv:1811.02584, Sec. 2.3.

[2] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3, Secs. 3.2–4.

[3] M. Josuat-Vergès and J. S. Kim, *Touchard–Riordan formulas, T-fractions, and Jacobi's triple product identity*, arXiv:1101.5608; FPSAC 2011 extended abstract, pp. 563–574.